where we have made the change of variables y2=1−d−x2. Continuing the string of equalities yields

=3 ∫ 0 1−2d (1−2d− x 1) 2 d x 1 =3 ∫ 0 1−2d y 1 2 d y 1 = (1−2d) 3

Hence, the desired probability that no 2 people are within a distance d of each other when 3 people are uniformly and independently distributed over an interval of size 1 is (1−2d)3 when d≤12. In fact, the same method can be used to prove that when n people are distributed at random over the unit interval, the desired probability is

[1−(n−1)d]n when d≤1n−1

The proof is left as an exercise.

The density function of the jth-order statistic X(j) can be obtained either by integrating the joint density function (**6.1**) or by direct reasoning as follows: In order for X(j) to equal x, it is necessary for j−1 of the n values X1,… ,Xn to be less than x,n−j of them to be greater than x, and 1 of them to equal x. Now, the probability density that any given set of j−1 of the Xi‘s are less than x, another given set of n−j are all greater than x, and the remaining value is equal to x equals

[F(x)]j−1[1−F(x)]n−jf(x)

Hence, since there are

(nj−1,n−j,1)=n!(n−j)!(j−1)!

different partitions of the n random variables X1,… ,Xn into the preceding three groups, it follows that the density function of X(j) is given by

fX(j)(x)=n!(n−j)!(j−1)![F(x)]j−1[1−F(x)]n−jf(x) (6.2)

Example 6b

When a sample of 2n+1 random variables (that is, when 2n+1 independent and identically distributed random variables) is observed, the (n+1) smallest is called the *sample median*. If a sample of size 3 from a uniform distribution over (0, 1) is observed, find the probability that the sample median is between 14 and 34.

Solution

From **Equation (6.2)**, the density of X(2) is given by

fX(2)(x)=3!1!1!x(1−x) 0<x<1

Hence,

P{ 1 4 < X (2) < 3 4 } =6 ∫ 1/4 3/4 x(1−x) dx =6{ x 2 2 − x 3 3 } | x=1/4 x=3/4 = 11 16

The cumulative distribution function of X(j) can be found by integrating **Equation (6.2)**. That is,

F X (j) (y)= n! (n−j)!(j−1)! ∫ −∞ y [F(x)] j−1 [1−F(x)] n−j f(x) dx (6.3)

However, FX(j)(y) could also have been derived directly by noting that the jth order statistic is less than or equal to y if and only if there are j or more of the Xi‘s that are less than or equal to y. Thus, because the number of Xi‘s that are less than or equal to y is a binomial random variable with parameters n,p=F(y), it

follows that

$$F_{X(j)}(y) = P\{X(j) \le y\} = P\{j \text{ or more of the } X_j\text{'s are} \le y\} = \sum_{k=j}^{n} \binom{n}{k} [F(y)]^k [1-F(y)]^{n-k} \tag{6.4}$$

If, in **Equations (6.3)** and **(6.4)**, we take F to be the uniform (0, 1) distribution [that is, f(x)=1,0<x<1], then we obtain the interesting analytical identity

$$\sum_{k=j}^{n} \binom{n}{k} y^k (1-y)^{n-k} = n!\,(n-j)!(j-1)! \int_0^y x^{j-1}(1-x)^{n-j}\,dx \quad 0 \le y \le 1 \tag{6.5}$$

We can employ the same type of argument that we used in establishing **Equation (6.2)** to find the joint density of X(i) and X(j), the ith and jth smallest of the values X1,… ,Xn. For suppose that i<j and xi<xj. Then the event that X(i)=xi,X(j)=xj is equivalent to the event that the n data values can be divided into 5 groups of respective sizes i−1, 1, j−i−1, 1, n−j, that satisfy the condition that all i−1 members of the first group have values less than xi, the one member of the second group has value xi, all j−i−1 members of the third group have values between xi and xj, the one member of the fourth group has value xj, and all n−j members of the last group have values greater than xj. Now, for any specified division of the n values into 5 such groups, the preceding condition will hold with probability (density)

$$F^{i-1}(x_i) f(x_i) (F(x_j)-F(x_i))^{j-i-1} f(x_j) (1-F(x_j))^{n-j}$$

As there are n!(i−1)!1!(j−i−1)!1!(n−j)! such divisions of the n values, and as the condition cannot hold for more than one of these divisions, it follows, for i<j, xi<xj, that

$$f_{X(i),X(j)}(x_i, x_j) = n!\,(i-1)!(j-i-1)!(n-j)!\, F^{i-1}(x_i) f(x_i) [F(x_j)-F(x_i)]^{j-i-1} f(x_j) [1-F(x_j)]^{n-j} \tag{6.6}$$

Example 6c Distribution of the range of a random sample

Suppose that n independent and identically distributed random variables X1,X2,… ,Xn are observed. The random variable R defined by R=X(n)−X(1) is called the *range* of the observed random variables. If the random variables Xi have distribution function F and density function f, then the distribution of R can be obtained from **Equation (6.6)** as follows: For a≥0,

$$P\{R \le a\} = P\{X(n) - X(1) \le a\} = \iint_{x_n - x_1 \le a} f_{X(1),X(n)}(x_1, x_n)\,dx_1\,dx_n = \int_{-\infty}^{\infty} \int_{-x_1}^{x_1+a} n!\,(n-2)! [F(x_n)-F(x_1)]^{n-2} f(x_1) f(x_n)\,dx_n\,dx_1$$

Making the change of variable y=F(xn)−F(x1),dy=f(xn) dxn yields

$$\int_{x_1}^{x_1+a} [F(x_n)-F(x_1)]^{n-2} f(x_n)\,dx_n = \int_0^{F(x_1+a)-F(x_1)} y^{n-2}\,dy = \frac{1}{n-1} [F(x_1+a)-F(x_1)]^{n-1}$$

Thus,

$$P\{R \le a\} = n \int_{-\infty}^{\infty} [F(x_1+a)-F(x_1)]^{n-1} f(x_1)\,dx_1 \tag{6.7}$$

Equation (6.7) can be evaluated explicitly only in a few special cases. One such case is when the Xi's are all uniformly distributed on (0, 1). In this case, we obtain, from **Equation (6.7)**, that for 0<a<1,

$$P\{R<a\} = n \int_0^1 [F(x_1+a)-F(x_1)]^{n-1} f(x_1)\,dx_1 = n \int_0^{1-a} a^{n-1}\,dx_1 + n \int_{1-a}^1 (1-x_1)^{n-1}\,dx_1 = n(1-a)a^{n-1} + a^n$$

Differentiation yields the density function of the range: given in this case by

$$f_R(a) = \begin{cases} n(n-1)a^{n-2}(1-a) & 0 \le a \le 1 \\ 0 & \text{otherwise} \end{cases}$$

That is, the range of n independent uniform (0, 1) random variables is a beta random variable with parameters n−1,2.

6.7 Joint Probability Distribution of Functions of Random Variables

Let X1 and X2 be jointly continuous random variables with joint probability density function fX1,X2. It is sometimes necessary to obtain the joint distribution of the random variables Y1 and Y2, which arise as functions of X1 and X2. Specifically, suppose that Y1=g1(X1,X2) and Y2=g2(X1,X2) for some functions g1 and g2.

Assume that the functions g1 and g2 satisfy the following conditions:

1. The equations y1=g1(x1,x2) and y2=g2(x1,x2) can be uniquely solved for x1 and x2 in terms of y1 and y2, with solutions given by, say, x1=h1(y1,y2),x2=h2(y1, y2).
2. The functions g1 and g2 have continuous partial derivatives at all points (x1,x2) and are such that the 2×2 determinant
   $J(x_1, x_2)=\begin{vmatrix} \partial g_1/\partial x_1 & \partial g_1/\partial x_2 \\ \partial g_2/\partial x_1 & \partial g_2/\partial x_2 \end{vmatrix} \equiv \partial g_1/\partial x_1 \, \partial g_2/\partial x_2 - \partial g_1/\partial x_2 \, \partial g_2/\partial x_1 \neq 0$

   at all points (x1,x2).

Under these two conditions, it can be shown that the random variables Y1 and Y2 are jointly continuous with joint density function given by

$$f_{Y_1Y_2}(y_1,y_2)=f_{X_1,X_2}(x_1,x_2)|J(x_1,x_2)|^{-1} \tag{7.1}$$

where x1=h1(y1, y2), x2=h2(y1,y2).

A proof of **Equation (7.1)** would proceed along the following lines:

$$P\{Y_1 \le y_1, Y_2 \le y_2\}=\iint_{(x_1,x_2):\, g_1(x_1,x_2)\le y_1,\, g_2(x_1,x_2)\le y_2} f_{X_1,X_2}(x_1,x_2)\,dx_1\,dx_2 \tag{7.2}$$

The joint density function can now be obtained by differentiating **Equation (7.2)** with respect to y1 and y2. That the result of this differentiation will be equal to the right-hand side of **Equation (7.1)** is an exercise in advanced calculus whose proof will not be presented in this book.

Example 7a

Let X1 and X2 be jointly continuous random variables with probability density function fX1,X2. Let Y1=X1+X2,Y2=X1−X2. Find the joint density function of Y1 and Y2 in terms of fX1,X2.

Solution

Let g1(x1,x2)=x1+x2 and g2(x1,x2)=x1−x2. Then

J(x1,x2)=|111−1|=−2

Also, since the equations y1=x1+x2 and y2=x1−x2 have x1=(y1+y2)/2, x2=(y1−y2)/2 as their solution, it follows from **Equation (7.1)** that the desired density is

fY1,Y2(y1,y2)=12fX1, X2(y1+y22,y1−y22)

For instance, if X1 and X2 are independent uniform (0, 1) random variables, then

$$f_{Y_1,Y_2}(y_1,y_2)=\begin{cases} \frac{1}{2} & 0\le y_1+y_2\le 2,\ 0\le y_1-y_2\le 2 \\ 0 & \text{otherwise} \end{cases}$$

or if X1 and X2 are independent exponential random variables with respective parameters λ1 and λ2, then

f Y 1 , Y 2 (y 1 , y 2) ={ λ 1 λ 2 2 exp{ − λ 1 (y 1 + y 2 2)− λ 2 (y 1 − y 2 2) } y 1 + y 2 ≥0, y 1 − y 2 ≥0 otherwise

Finally, if X1 and X2 are independent standard normal random variables, then

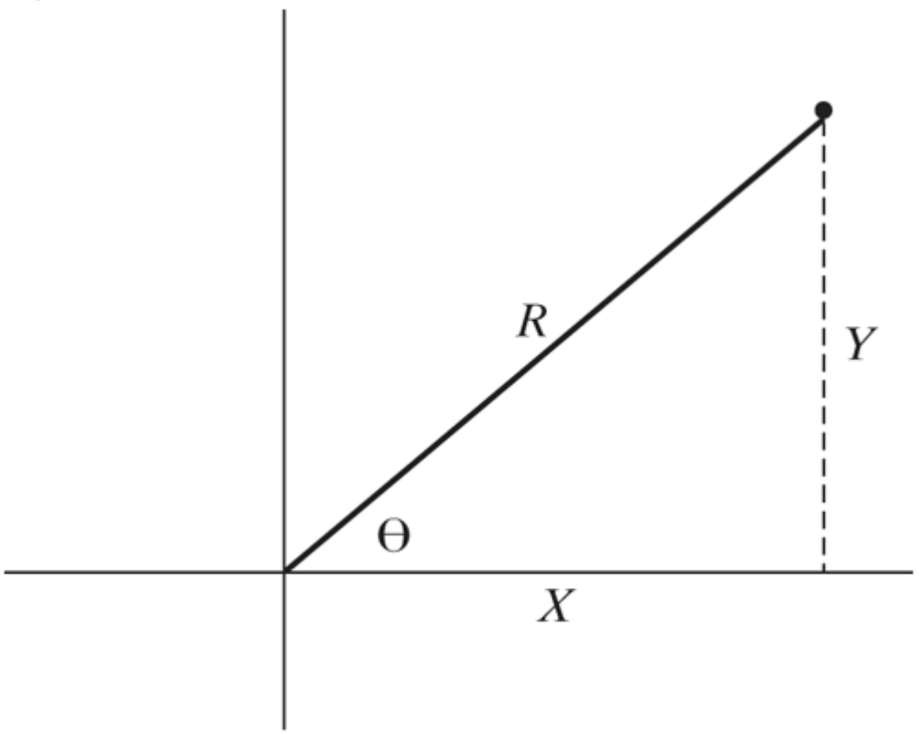
f Y 1 , Y 2 (y 1 , y 2) = 1 4π e −[(y 1 + y 2) 2 /8+ (y 1 − y 2) 2 /8] = 1 4π e −(y 2 1 + y 2 2)/4 = 1 4π e − y 2 1 /4 1 4π e − y 2 2 /4

Thus, not only do we obtain (in agreement with **Proposition 3.2**) that both X1+X2 and X1−X2 are normal with mean 0 and variance 2, but we also conclude that these two random variables are independent. (In fact, it can be shown that if X1 and X2 are independent random variables having a common distribution function F, then X1+X2 will be independent of X1−X2 if and only if F is a normal distribution function.)

Example 7b

Let (X, Y) denote a random point in the plane, and assume that the rectangular coordinates X and Y are independent standard normal random variables. We are interested in the joint distribution of R, Θ, the polar coordinate representation of (x, y). (See **Figure 6.4**.)

Figure 6.4 •= Random point. (X,Y)=(R, Θ).



Suppose first that X and Y are both positive. For x and y positive, letting r=g1(x,y)=x2+y2 and θ=g2(x,y)=tan−1y/x, we see that

∂ g 1 ∂x = x x 2 + y 2 ∂ g 1 ∂y = y x 2 + y 2 ∂ g 2 ∂x = 1 1+ (y/x) 2 (−y x 2)= −y x 2 + y 2 ∂ g 2 ∂y = 1 x[1+ (y/x) 2] = x x 2 + y 2

Hence,

J(x,y)=x2(x2+y2)3/2+y2(x2+y2)3/2=1x2+y2=1r

Because the conditional joint density function of X, Y given that they are both positive is

f(x,y|X>0,Y>0)=f(x,y)P(X>0,Y>0)=2πe−(x2+y2)/2, x>0,y>0

we see that the conditional joint density function of R=X2+Y2 and Θ=tan−1(Y/X), given that X and Y are both positive, is

f(r,θ|X>0,Y>0)= 2 π r e − r 2/2 , 0<θ<π/2, 0<r<∞

Similarly, we can show that

f(r,θ|X<0,Y>0)= 2 π r e − r 2/2 , π/2<θ<π, 0<r<∞ f(r,θ|X<0,Y<0)= 2 π r e − r 2/2 , π<θ<3π/2, 0<r<∞
f(r,θ|X>0,Y<0)= 2 π r e − r 2/2 , 3π/2<θ<2π, 0<r<∞

As the joint density is an equally weighted average of these four conditional joint densities, we obtain that the joint density of R,Θ is given by

f(r,θ)=12πre−r2/2 0<θ<2π, 0<r<∞

Now, this joint density factors into the marginal densities for R and Θ, so R and Θ are independent random variables, with Θ being uniformly distributed over (0,2π) and R having the Rayleigh distribution with density

f(r)=re−r2/2 0<r<∞

(For instance, when one is aiming at a target in the plane, if the horizontal and vertical miss distances are independent standard normals, then the absolute value of the error has the preceding Rayleigh distribution.)

This result is quite interesting, for it certainly is not evident a priori that a random vector whose coordinates are independent standard normal random variables will have an angle of orientation that not only is uniformly distributed, but also is independent of the vector's distance from the origin.

If we wanted the joint distribution of R2 and Θ, then, since the transformation d=g1(x,y)=x2+y2 and θ=g2(x,y)=tan−1y/x has the Jacobian

J=| 2x −y x 2 + y 2 2y x x 2 + y 2 |=2

it follows that

f(d,θ)=12e−d/212π 0<d<∞, 0<θ<2π

Therefore, R2 and Θ are independent, with R2 having an exponential distribution with parameter 12. But because R2=X2+Y2, it follows by definition that R2 has a chi-squared distribution with 2 degrees of freedom. Hence, we have a verification of the result that the exponential distribution with parameter 12 is the same as the chi-squared distribution with 2 degrees of freedom.

The preceding result can be used to simulate (or generate) normal random variables by making a suitable transformation on uniform random variables. Let U1 and U2 be independent random variables, each uniformly distributed over (0, 1). We will transform U1,U2 into two independent standard normal random variables X1 and X2 by first considering the polar coordinate representation (R,Θ) of the random vector (X1,X2). From the preceding, R2 and Θ will be independent, and, in addition,

R2=X21+X22 will have an exponential distribution with parameter λ=12. But −2 log U1 has such a distribution, since, for x>0,

$$P\{-2\log U_1 < x\} = P\{\log U_1 > -x/2\} = P\{U_1 > e^{-x/2}\} = 1 - e^{-x/2}$$

Also, because 2π U2 is a uniform (0,2π) random variable, we can use it to generate Θ. That is, if we let

$$R^2 = -2\log U_1 \quad \Theta = 2\pi U_2$$

then R2 can be taken to be the square of the distance from the origin and θ can be taken to be the angle of orientation of (X1,X2). Now, since X1=R cos Θ,X2=R sin Θ, it follows that

$$X_1 = \sqrt{-2\log U_1}\cos(2\pi U_2) \quad X_2 = \sqrt{-2\log U_1}\sin(2\pi U_2)$$

are independent standard normal random variables.

Example 7c

If X and Y are independent gamma random variables with parameters (α,λ) and (β,λ), respectively, compute the joint density of U=X+Y and V=X/(X+Y).

Solution

The joint density of X and Y is given by

$$f_{X,Y}(x,y) = \lambda e^{-\lambda x}(\lambda x)^{\alpha-1}\Gamma(\alpha)\,\lambda e^{-\lambda y}(\lambda y)^{\beta-1}\Gamma(\beta) = \lambda^{\alpha+\beta}\Gamma(\alpha)\Gamma(\beta)\, e^{-\lambda(x+y)}\, x^{\alpha-1} y^{\beta-1}$$

Now, if g1(x,y)=x+y,g2(x,y)=x/(x+y), then

$$\partial g_1/\partial x = \partial g_1/\partial y = 1 \quad \partial g_2/\partial x = y/(x+y)^2 \quad \partial g_2/\partial y = -x/(x+y)^2$$

so

$$J(x,y) = \begin{vmatrix} 1 & y/(x+y)^2 \\ 1 & -x/(x+y)^2 \end{vmatrix} = -\frac{1}{x+y}$$

Finally, as the equations u=x+y,v=x/(x+y) have as their solutions x=uv,y=u(1−v), we see that

$$f_{U,V}(u,v) = f_{X,Y}[uv, u(1-v)]u = \lambda e^{-\lambda u}(\lambda u)^{\alpha+\beta-1}\Gamma(\alpha+\beta)\, v^{\alpha-1}(1-v)^{\beta-1}\Gamma(\alpha+\beta)\Gamma(\alpha)\Gamma(\beta)$$

Hence, X+Y and X/(X+Y) are independent, with X+Y having a gamma distribution with parameters (α+β,λ) and X/(X+Y) having a beta distribution with parameters (α,β). The preceding reasoning also shows that B(α,β), the normalizing factor in the beta density, is such that

$$B(\alpha,\beta) \equiv \int_0^1 v^{\alpha-1}(1-v)^{\beta-1}\,dv = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$$

This entire result is quite interesting. For suppose there are n+m jobs to be performed, each (independently) taking an exponential amount of time with rate λ to be completed and suppose that we have two workers to perform these jobs. Worker I will do jobs 1,2,… ,n, and worker II will do the remaining m jobs. If we let X and Y denote the total working times of workers I and II, respectively, then (either from the foregoing result or from **Example 3b**) X and Y will be independent gamma random variables having parameters (n,λ) and (m,λ), respectively. It then follows that independently of the working time needed to complete all n+m jobs (that is, of X+Y), the proportion of this work that will be performed by worker I has a beta distribution with parameters (n, m).

When the joint density function of the n random variables X1,X2,… ,Xn is given and we want to compute the joint density function of Y1,Y2,… ,Yn, where

Y1=g1(X1,… ,Xn) Y2=g2(X1,… ,Xn),… Yn=gn(X1,… ,Xn)

the approach is the same—namely, we assume that the functions gi have continuous partial derivatives and that the Jacobian determinant

J(x1,… ,xn)=|∂g1∂x1∂g1∂x2⋯∂g1∂xn∂g2∂x1∂g2∂x2⋯∂g2∂xn∂gn∂x1∂gn∂x2⋯∂gn∂xn|≠0

at all points (x1,… ,xn). Furthermore, we suppose that the equations y1=g1(x1,… ,xn),y2=g2(x1,… ,xn),… ,yn=gn(x1,… ,xn) have a unique solution, say, x1=h1(y1,… ,yn),… ,xn=hn(y1,… ,yn). Under these assumptions, the joint density function of the random variables Yi is given by

fY1,… ,Yn(y1,… ,yn)=fX1,… ,Xn(x1,… ,xn)|J(x1,… ,xn)|−1 (7.3)

where xi=hi(y1,… ,yn),i=1,2,… ,n.

Example 7d

Let X1,X2, and X3 be independent standard normal random variables. If Y1=X1+X2+X3,Y2=X1−X2, and Y3=X1−X3, compute the joint density function of Y1,Y2,Y3.

Solution

Letting Y1=X1+X2+X3,Y2=X1−X2,Y3=X1−X3, the Jacobian of these transformations is given by

J=|1111−1010−1|=3

As the preceding transformations yield that

X1=Y1+Y2+Y33 X2=Y1−2Y2+Y33 X3=Y1+Y2−2Y33

we see from **Equation (7.3)** that

f Y 1 , Y 2 , Y 3 (y 1 , y 2 , y 3) = 1 3 f X 1 , X 2 , X 3 (y 1 + y 2 + y 3 3 , y 1 −2 y 2 + y 3 3 , y 1 + y 2 −2 y 3 3)

Hence, as

f X 1 , X 2 , X 3 (x 1 , x 2 , x 3)= 1 (2π) 3/2 e − ∑ i=1 3 x 2 i /2

we see that

fY1, Y2, Y3(y1, y2, y3)=13(2π)3/2e−Q(y1, y2, y3)/2

where

Q(y 1 , y 2 , y 3) = (y 1 + y 2 + y 3 3) 2 + (y 1 −2 y 2 + y 3 3) 2 + (y 1 + y 2 −2 y 3 3) 2 = y 2 1 3 + 2 3 y 2 2 + 2 3 y 2 3 − 2 3 y 2 y 3

Example 7e

Let X1,X2,… ,Xn be independent and identically distributed exponential random variables with rate λ. Let

Yi=X1+⋯+Xi i=1,… ,n

a. Find the joint density function of Y1,… ,Yn.
b. Use the result of part (a) to find the density of Yn.
c. Find the conditional density of Y1,… ,Yn−1 given that Yn=t.

Solution

a. The Jacobian of the transformations Y1=X1, Y2=X1+X2, … , Yn=X1+⋯+Xn is
J=| 1 0 0 0 ⋯ 0 1 1 0 0 ⋯ 0 1 1 1 0 ⋯ 0 ⋯ ⋯ ⋯ ⋯ 1 1 1 1 ⋯ 1 |

   Since only the first term of the determinant will be nonzero, we have J=1. Now, the joint density function of X1,… ,Xn is given by
fX1,… ,Xn(x1,… ,xn)=∏i=1nλe−λxi 0<xi<∞, i=1,… ,n

   Hence, because the preceding transformations yield
X1=Y1,X2=Y2−Y1,… ,Xi=Yi−Yi−1,… ,Xn=Yn−Yn−1

   it follows from **Equation (7.3)** that the joint density function of Y1,… ,Yn is
f Y 1 ,… , Y n (y 1 ,… , y n)= f X 1 ,… , X n (y 1 , y 2 − y 1 ,… , y n − y n−1) = λ n exp { −λ [y 1 + ∑ i=2 n (y i − y i−1)] } = λ n e −λ y n 0< y 1 ,0< y i − y i−1 ,i=2,… ,n = λ n e −λ y n 0< y 1 < y 2 <⋯< y n

b. To obtain the marginal density of Yn, let us integrate out the other variables one at a time. Doing this gives
f Y 2 ,… , Y n (y 2 ,… , y n) = ∫ 0 y 2 λ n e −λ y n d y 1 = λ n y 2 e −λ y n 0< y 2 < y 3 <⋯< y n

   Continuing, we obtain
f Y 3 ,… , Y n (y 3 ,… , y n) = ∫ 0 y 3 λ n y2 e −λ y n d y 2 = λ n y 3 2 2 e −λ y n 0< y 3 < y 4 <⋯< y n

   The next integration yields
f Y 4 ,… , Y n (y 4 ,… , y n)= λ n y 4 3 3! e −λ y n 0< y 4 <⋯< y n

   Continuing in this fashion gives
f Y n (y n)= λ n y n n−1 (n−1)! e −λ y n 0< y n

   which, in agreement with the result obtained in **Example 3b**, shows that X1+⋯+Xn is a gamma random variable with parameters n and λ.
c. The conditional density of Y1,… ,Yn−1 given that Yn=t is, for 0<y1<… <yn−1<t,
f Y 1 ,… , Y n−1 | Y n (y 1 ,… , y n−1 |t) = f Y 1 ,… , Y n−1 , Y n (y 1 ,… , y n−1 ,t) f Y n (t) = λ n e −λt λ e −λt (λt) n−1 /(n−1)! = (n−1)! t n−1

Because f(y)=1/t, 0<y<t, is the density of a uniform random variable on (0,t), it follows that conditional on Yn=t, Y1,… ,Yn−1 are distributed as the order statistics of n−1 independent uniform (0,t) random variables.

* 6.8 Exchangeable Random Variables

The random variables $X_1, X_2, \ldots, X_n$ are said to be *exchangeable* if, for every permutation $i_1, \ldots, i_n$ of the integers $1, \ldots, n$,

$$P\{X_{i_1} \le x_1, X_{i_2} \le x_2, \ldots, X_{i_n} \le x_n\} = P\{X_1 \le x_1, X_2 \le x_2, \ldots, X_n \le x_n\}$$

for all $x_1, \ldots, x_n$. That is, the n random variables are exchangeable if their joint distribution is the same no matter in which order the variables are observed.

Discrete random variables will be exchangeable if

$$P\{X_{i_1} = x_1, X_{i_2} = x_2, \ldots, X_{i_n} = x_n\} = P\{X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n\}$$

for all permutations $i_1, \ldots, i_n$, and all values $x_1, \ldots, x_n$. This is equivalent to stating that $p(x_1, x_2, \ldots, x_n) = P\{X_1 = x_1, \ldots, X_n = x_n\}$ is a symmetric function of the vector $(x_1, \ldots, x_n)$, which means that its value does not change when the values of the vector are permuted.

Example 8a

Suppose that balls are withdrawn one at a time and without replacement from an urn that initially contains n balls, of which k are considered special, in such a manner that each withdrawal is equally likely to be any of the balls that remains in the urn at the time. Let $X_i = 1$ if the ith ball withdrawn is special and let $X_i = 0$ otherwise. We will show that the random variables $X_1, \ldots, X_n$ are exchangeable. To do so, let $(x_1, \ldots, x_n)$ be a vector consisting of k ones and $n-k$ zeros. However, before considering the joint mass function evaluated at $(x_1, \ldots, x_n)$, let us try to gain some insight by considering a fixed such vector for instance, consider the vector $(1,1,0,1,0,\ldots,0,1)$, which is assumed to have k ones and $n-k$ zeros. Then

$$p(1,1,0,1,0,\ldots,0,1) = \frac{k}{n}\frac{k-1}{n-1}\frac{n-k}{n-2}\frac{k-2}{n-3}\frac{n-k-1}{n-4}\cdots\frac{1}{1}$$

which follows because the probability that the first ball is special is k/n, the conditional probability that the next one is special is $(k-1)/(n-1)$, the conditional probability that the next one is not special is $(n-k)/(n-2)$, and so on. By the same argument, it follows that $p(x_1, \ldots, x_n)$ can be expressed as the product of n fractions. The successive denominator terms of these fractions will go from n down to 1. The numerator term at the location where the vector $(x_1, \ldots, x_n)$ is 1 for the ith time is $k-(i-1)$, and where it is 0 for the ith time it is $n-k-(i-1)$. Hence, since the vector $(x_1, \ldots, x_n)$ consists of k ones and $n-k$ zeros, we obtain

$$p(x_1, \ldots, x_n) = \frac{k!(n-k)!}{n!} \quad x_i = 0,1, \sum_{i=1}^{n} x_i = k$$

Since this is a symmetric function of $(x_1, \ldots, x_n)$, it follows that the random variables are exchangeable.

Remark Another way to obtain the preceding formula for the joint probability mass function is to regard all the n balls as distinguishable from one another. Then, since the outcome of the experiment is an ordering of these balls, it follows that there are $n!$ equally likely outcomes. Finally, because the number of outcomes having special and nonspecial balls in specified places is equal to the number of ways of permuting the special and the nonspecial balls among themselves, namely $k!(n-k)!$, we obtain the preceding mass function.

It is easily seen that if $X_1, X_2, \ldots, X_n$ are exchangeable, then each X_i has the same probability distribution. For instance, if X and Y are exchangeable discrete random variables, then

$$P\{X = x\} = \sum_y P\{X = x, Y = y\} = \sum_y P\{X = y, Y = x\} = P\{Y = x\}$$

For example, it follows from **Example 8a** that the ith ball withdrawn will be special with probability k/n, which is intuitively clear, since each of the n balls is equally likely to be the ith one selected.

Example 8b

In **Example 8a**, let Y1 denote the selection number of the first special ball withdrawn, let Y2 denote the additional number of balls that are then withdrawn until the second special ball appears, and, in general, let Yi denote the additional number of balls withdrawn after the (i−1) special ball is selected until the ith is selected, i=1,… ,k. For instance, if n=4,k=2 and X1=1,X2=0,X3=0,X4=1, then Y1=1,Y2=3. Now, Y1=i1,Y2=i2,… ,Yk=ik⇔Xi1=Xi1+i2=⋯=Xi1+⋯+ik=1,Xj=0, otherwise; thus, from the joint mass function of the Xi, we obtain

$$P\{Y1=i1,Y2=i2,\dots,Yk=ik\}=k!(n-k)!n! \quad i1+\cdots+ik\le n$$

Hence, the random variables Y1,… ,Yk are exchangeable. Note that it follows from this result that the number of cards one must select from a well-shuffled deck until an ace appears has the same distribution as the number of additional cards one must select after the first ace appears until the next one does, and so on.

Example 8c

The following is known as Polya's urn model: Suppose that an urn initially contains n red and m blue balls. At each stage, a ball is randomly chosen, its color is noted, and it is then replaced along with another ball of the same color. Let Xi=1 if the ith ball selected is red and let it equal 0 if the ith ball is blue, i≥1. To obtain a feeling for the joint probabilities of these Xi, note the following special cases:

$$P\{X_1=1, X_2=1, X_3=0, X_4=1, X_5=0\} = \frac{n}{n+m}\,\frac{n+1}{n+m+1}\,\frac{m}{n+m+2}\,\frac{n+2}{n+m+3}\,\frac{m+1}{n+m+4}$$
$$= \frac{n(n+1)(n+2)m(m+1)}{(n+m)(n+m+1)(n+m+2)(n+m+3)(n+m+4)}$$

and

$$P\{X_1=0, X_2=1, X_3=0, X_4=1, X_5=1\} = \frac{m}{n+m}\,\frac{n}{n+m+1}\,\frac{m+1}{n+m+2}\,\frac{n+1}{n+m+3}\,\frac{n+2}{n+m+4}$$
$$= \frac{n(n+1)(n+2)m(m+1)}{(n+m)(n+m+1)(n+m+2)(n+m+3)(n+m+4)}$$

By the same reasoning, for any sequence x1,… ,xk that contains r ones and k−r zeros, we have

$$P\{X_1=x_1,\dots,X_k=x_k\} = \frac{n(n+1)\cdots(n+r-1)m(m+1)\cdots(m+k-r-1)}{(n+m)\cdots(n+m+k-1)}$$

Therefore, for any value of k, the random variables X1,… ,Xk are exchangeable.

An interesting corollary of the exchangeability in this model is that the probability that the ith ball selected is red is the same as the probability that the first ball selected is red, namely, $\frac{n}{n+m}$. (For an intuitive argument for this initially nonintuitive result, imagine that all the n+m balls initially in the urn are of different types. That is, one is a red ball of type 1, one is a red ball of type 2, … , one is a red ball type of n, one is a blue ball of type 1, and so on, down to the blue ball of type m. Suppose that when a ball is selected it is replaced along with another of its type. Then, by symmetry, the ith ball selected is equally likely to be of any of the n+m distinct types. Because n of these n+m types are red, the probability is $\frac{n}{n+m}$.)

Our final example deals with continuous random variables that are exchangeable.

Example 8d

Let X1,X2,… ,Xn be independent uniform (0, 1) random variables, and denote their order statistics by